

A12749W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics

Honour School of Economics and Management

Honour School of History and Economics

Honour School of Engineering, Economics and Management (Part II)

Honour School of Materials, Economics and Management (Part I)

MICROECONOMICS

OUTLINE ANSWERS TO PROBLEMS

TRINITY TERM 2016

Friday 27th May 2016, 09:30 - 12:30

Please start the answer to each question on a separate page.

There are 10 questions in this paper.

*Answer **all** questions in Part A and **two** long questions from Part B.*

Part A attracts 1/3 of the marks in total; each question in Part B attracts 1/3 of the marks.

For the questions in Part A, and the multipart question in Part B, the weight assigned to each part is indicated in square brackets.

Candidates may use their own calculators.

Do not turn over until told that you may do so.

Part A

Numbers in square brackets indicate marks as a percentage of the total for Part A

1. "When firms compete in prices, prices fall to marginal cost and the firms make no profits."
 - (a) Describe carefully a game-theoretic model of a price-setting duopoly that illustrates this statement. [10%]

The 2 firms produce homogeneous goods; both have marginal cost c .

The firms simultaneously set prices: p_1, p_2 .

Market demand is $q(p)$.

If $p_i < p_j$, all demand goes to firm i : $q_i = q(p_i)$ and $q_j = 0$;

if $p_i = p_j$, each firm gets half the market: $q_i = q_j = q(p_i)/2$.

The Nash equilibrium is $p_1 = p_2 = c$; since every good is sold at unit cost, profit is zero.

Could illustrate by drawing best responses. Alternatively, argue verbally:

To show that this is an equilibrium: if firm i lowered its price its profit would be negative; if it raised its price, profit would still be zero.

To show that it is the only pure strategy equilibrium, and illuminate the statement: clearly it can never be a best response for i to set $p_i > p_j$,

or $p_i < c$. $p_i = p_j > c$ is not a BR either, because each firm would want to undercut the other: i would prefer $p_i = p_j - \varepsilon$.

- (b) Briefly describe a modification of your model in which the statement would not hold. Explain intuitively why in this case firms are able to make profits in equilibrium. [10%]

Perhaps most obvious modification would be differentiated products;

this could be explained by specifying firm demand as $q_i = a - bp_i + gp_j$

where a, b, g are positive and $b > g$. Not necessary to derive best responses, although they might be drawn. Intuition is more important --

how differentiation changes firm demand, less intense competition, etc.

Equally acceptable alternatives (all mentioned in lectures):

mixed strategy equilibrium, collusion in repeated interaction,

capacity constraints, and different marginal costs (the easiest to explain).

2. (a) Define the risk premium associated with a lottery. [5%]

The certainty equivalent is y^c where $u(y^c) = \bar{u}$
and $\bar{u}$ is the expected value of the lottery.

Let $\bar{y}$ be the expected outcome of the lottery.

The risk premium $r = \bar{y} - y^c$,

the amount the agent would pay to remove the risk
(or would require to accept the risk).

Janet has a ticket for a lottery that will leave her with final wealth of either 10 or 100, each equally likely; there is another lottery that (if she owned a ticket) would leave Janet with final wealth of either 35 or 65, again each equally likely, but it is Sam that has a ticket for this second lottery.

- (b) If Janet is willing to exchange tickets with Sam, what does this say about Janet's attitude to risk? [9%]

A rather rough argument, but enough for most of the marks, would be:

For J, L_1 has lower expected utility than L_2 : $\frac{1}{2}u(10) + \frac{1}{2}u(100) \leq \frac{1}{2}u(35) + \frac{1}{2}u(65)$
 L_1 has higher mean ($55 > 50$), but a spread distribution ($10 < 35$ & $100 > 65$).

Since J is prepared to accept a lower mean for a less spread distribution, she must be risk averse.

{NOTE >>>} An argument just using 'variance' instead of 'spread' is invalid.

A precise argument can be made by comparing the two lotteries with a third:

For Janet: $\frac{1}{2}u(10) + \frac{1}{2}u(100) \leq \frac{1}{2}u(35) + \frac{1}{2}u(65)$.

For an EU maximizer:

$\frac{1}{2}u(35) + \frac{1}{2}u(65) < \frac{1}{2}u(40) + \frac{1}{2}u(70)$ (higher mean, same variance);

if risk neutral: $\frac{1}{2}u(40) + \frac{1}{2}u(70) = \frac{1}{2}u(10) + \frac{1}{2}u(100)$ (same mean);

if risk loving: $\frac{1}{2}u(40) + \frac{1}{2}u(70) < \frac{1}{2}u(10) + \frac{1}{2}u(100)$ (mean preserving spread).

So Janet cannot be either risk neutral or risk loving -- she is risk averse.

- (c) Sam does not agree to a straight exchange of tickets, but requires Janet to pay a price p in addition. If Janet derives utility $\ln w$ from final wealth w , what is the highest value of p that she is prepared to pay? [9%]

To make J indifferent: $\frac{1}{2}u(35 - p) + \frac{1}{2}u(65 - p) = \frac{1}{2}u(10) + \frac{1}{2}u(100)$

With $u(w) = \ln w$:

$$\begin{aligned} (35 - p)(65 - p) &= 1000 \\ \Rightarrow p^2 - 100p + 1275 &= 0 \\ p &= \frac{100 \pm \sqrt{10000 - 5100}}{2} = \frac{100 \pm \sqrt{4900}}{2} = \frac{100 \pm 70}{2} \\ p &= 15 \end{aligned}$$

3. Three bicycles are available: one each of high, medium and low quality. There are three type *X* consumers, two type *Y* consumers and two type *Z* consumers where each consumer-type values each quality of bicycle differently:

Quality	Type <i>X</i> value	Type <i>Y</i> value	Type <i>Z</i> value
<i>High</i>	90	100	115
<i>Medium</i>	35	60	50
<i>Low</i>	25	20	30

Everyone knows the value that each type of consumer attaches to each quality of bicycle. No-one wants more than one bicycle, and the price at which any bicycle is traded will be determined by supply and demand.

- (a) Suppose that type *X* consumers initially own the bicycles – one each. If *all* the consumers know the quality of each bicycle, which consumer-type will end up owning which bicycle?

Is the equilibrium price for each bike fully determined? [8%]

The H and L bikes will be owned by Zs, and the M bike will be owned by a Y. Prices are not fully determined.

Ys outbid Zs for the M bike.

Competition between Ys bids the price of M up to 60.

Zs outbid Ys for the H and L bikes. Prices indeterminate.

If one of these bikes is traded first, for example, it will be sold at the valuation of a Z. The other Z can buy the other bike at a price between the Z valuation and the next highest.

From now on, assume that buyers do not know the quality of any particular bicycle for sale, but sellers do know the quality of the one that they own.

- (b) Continue to suppose that type *X* consumers initially have one bicycle each. Is there adverse selection in this market? What will be the equilibrium outcome? [10%]

Let p be the market price.

If $p \geq 90$, supply is 3, but the average value of a bike is 60 to Ys and 65 to Zs. No-one will buy.

If $90 > p \geq 35$ supply is 2.

H is not traded, so the expected value of a bike is 40 to Ys and Zs.

Demand is 4; the market price is 40, and both M and L bikes are sold.

Yes, there is Adverse Selection.

{ADDED LATER >>>} Note that there is a second equilibrium outcome:

If $35 > p \geq 25$ supply is 1. Value is 20 to Ys and 30 to Zs; so demand is 2.

The market price is 30, and the L bike is sold to a Z.

- (c) The market in part (b) has closed, and any new owner of a bicycle now knows the quality they bought. Suppose that a type Z consumer bought the low quality bicycle in that market. If the market were to reopen for a second round of trading, describe what the equilibrium outcome would be. [8%]

H is owned by an X, L by a Z, and M by a Y or a Z.

As before, no-one will pay 90, so H is not traded.

Then M can't be traded either; no-one would pay more than 40.

But L is already owned by the consumer who values it most. So no trade.

{ADDED LATER >>>} Note that in the second equilibrium outcome in part (b):

H is owned by an X, L by a Z, and now M is also still owned by an X.

This is essentially the same set-up that led to the first equilibrium outcome in part (b):

At a price of 40, X & Z would sell the M & L bikes,
and both the Ys & the other Z would buy.

We do not expect (many) students to identify this second equilibrium outcome, and is included here for completeness.

Do not penalise students that ignore it.

4. A company has purchased the rights to a very large number of music tracks, and incurs no further costs when tracks are downloaded and played. There is a large number, N , of consumers, who can play a track at price p . Consumers have quasi-linear utility: their net benefit from playing x tracks is

$$B(x) = ax - bx^2 - h - px$$

where a and b are parameters such that $a^2 < 4b$. Subscription is free, but h is the initial hassle (disutility) of subscribing to the service. Consumers differ in the level of hassle h that they face: h is uniformly distributed between 0 and 1.

- (a) Find the number of tracks, x^* , that a subscriber of type h will choose to play at price $p < a$, and the net benefit that the subscriber will obtain. [8%]

$$B'(x) = a - 2bx - p = 0 \implies x^* = \frac{a-p}{2b}, \quad B^*(p) = \frac{(a-p)^2}{4b} - h.$$

The SOC is satisfied: $B''(x) = -2b$.

- (b) Show that the number of consumers who will subscribe is $N^* = N \frac{(a-p)^2}{4b}$. [4%]

Consumers subscribe if $B^*(p) \geq 0$; that is, if $h \leq \frac{(a-p)^2}{4b}$, which is less than 1 for all $p \geq 0$. From the uniform distribution, $N^* = N \frac{(a-p)^2}{4b}$.

- (c) Show that the profit-maximising price for the firm is $p^* = a/4$. (There is no need to check the second order condition.) How many consumers will subscribe? [8%]

$$\pi = N^* x^* p = \frac{N}{8b^2} p (a-p)^3. \quad \text{Differentiating: } \frac{d\pi}{dp} = \frac{N}{8b^2} ((a-p)^3 - 3p(a-p)^2) = 0$$

$$\implies a-p = 3p \text{ (since } a=p \text{ cannot be a maximum) so } p^* = a/4.$$

$$\text{The equilibrium number of subscribers is } N^* = \frac{N}{4b} \left(\frac{3a}{4} \right)^2 = \frac{9a^2 N}{64b}$$

- (d) What would be the efficient price level, and how many consumers would subscribe at that price? [5%]

Efficiency requires price = marginal cost = 0.

$$\text{When } p=0 \text{ the number of subscribers is } N^e = \frac{a^2 N}{4b}$$

- (e) Discuss, without further calculations, the comparison between welfare (consumer and producer surplus) at the profit-maximising price level, and welfare at the efficient price level. [6%]

At the efficient price level, all benefit functions are shifted up, so there are more subscribers, and each subscriber has a higher net benefit. So total consumer surplus is higher at $p=0$ than at p^* .

The firm's surplus, which is positive at p^* , is zero at the efficient price level.

We know that total surplus is higher at $p=0$ -- obviously, because the efficient price maximises surplus.

But in addition we can say that the difference in total surplus is equal to the surplus of the original subscribers from the extra tracks they play, plus the surplus of the additional subscribers in the efficient case.

If the price falls from p^* to zero, then

- (i) the original N^* consumers still 'buy', and the firm's surplus on the tracks they used to buy, $p^*x^*N^*$, is transferred to them with no effect on total surplus, but they also play more tracks at the lower price; and
- (ii) additional consumers buy, and each of these receives a surplus.

Part B

10. Suppose that two firms engage in Cournot competition. The firms simultaneously choose individual production $q_i \geq 0$ ($i = 1, 2$). The market clears at a price $p = 100 - Q$, where $Q = q_1 + q_2$. Each firm has a cost function $C_i(q_i) = 10q_i$.

- (a) Find the unique pure-strategy Cournot-Nash equilibrium of this game.

How much profit does each firm earn in the equilibrium?

[15%]

$$\pi_i = (100 - q_i - q_j) q_i - 10q_i = (90 - q_i - q_j) q_i$$

$$\text{FOC: } \pi'_i = 90 - 2q_i - q_j = 0$$

$$\text{SOC: } \pi''_i = -2 < 0$$

$$\text{From FOC \& SOC, } q_i^* = \frac{90 - q_j}{2}$$

From FOC, $q_i^* = 90 - Q$ and $q_j^* = 90 - Q$, and so the equilibrium must be symmetric

Thus $q_i^* = \frac{90 - q_i^*}{2}$, which implies $q_i^* = 30$

$$\pi_i^* = (90 - 60) 30 = 30^2 = 900$$

Now suppose that before the two firms engage in Cournot competition, firm 1 can invest in cost reduction, but firm 2 cannot. If firm 1 invests, it incurs a cost of $K > 0$ and its cost function in the Cournot competition drops to $C_i(q_i) = 4q_i$. Firm 2 discovers whether firm 1 invested or not before Cournot competition begins.

- (b) If firm 1 chooses to invest, what will be the Cournot-Nash equilibrium of the quantity-setting game? How much profit does each firm earn in the equilibrium (ignoring any prior investment cost of firm 1)?

[15%]

$$\pi_1 = (100 - q_1 - q_2) q_1 - 4q_1 = (96 - q_1 - q_2) q_1$$

$$\text{FOC: } \pi'_1 = 96 - 2q_1 - q_2 = 0$$

$$\text{SOC: } \pi''_2 = -2 < 0$$

$$\text{From FOC \& SOC, } q_1^* = \frac{96 - q_2}{2}$$

$$\text{From part (a), } q_2^* = \frac{90 - q_1}{2}$$

Solving simultaneously gives $q_1^* = 34$ and $q_2^* = 28$

$$\pi_1^* = (96 - 34 - 28) \times 34 = 34^2 = 1156$$

$$\pi_2^* = (90 - 34 - 28) \times 28 = 28^2 = 784$$

- (c) Explain intuitively why firm 1 earns more than firm 2 in the Cournot-Nash equilibrium found in part (b). [8%]

Firm 1 earns more for two reasons.

First, its cost of production has fallen for any level of output.

Second, its lower production cost makes it more aggressive, pushing out its reaction curve for any q_2 .

This in turn reduces firm 2's production in equilibrium and increases firm 1's equilibrium production.

- (d) Consider a subgame-perfect Nash equilibrium of the two-stage game in which: (i) firm 1 chooses to invest or not in the first stage; and (ii) the firms engage in Cournot competition in the second stage after observing firm 1's investment decision in the first stage. For what values of K will firm 1 invest? Describe fully the strategies in the subgame-perfect Nash equilibrium in which firm 1 chooses to invest. [25%]

Solve by backward induction.

We already know the second-stage equilibrium after investment (part (b)) and after no investment (part (a)).

In the first stage, firm 1 will invest if and only if the increase in second-stage profits covers the investment cost.

The increase in profits is $1156 - 900 = 256$

And so there is a subgame-perfect Nash equilibrium in which firm 1 invests if and only if $K \leq 256$.

Strategy for firm 1:

- Invest in stage 1.

If invested in stage 1, set $q_1^* = 34$ in stage 2;

if did not invest in stage 1, set $q_1^* = 30$ in stage 2.

Strategy for firm 2:

- If firm 1 invested in stage 1, set $q_2^* = 28$ in stage 2;

if firm 1 did not invest in stage 1, set $q_2^* = 30$ in stage 2.

- (e) Suppose that $K = 200$. Show that there is a Nash equilibrium of the two-stage game in which firm 1 does not invest in the first stage and neither firm conditions its production level on whether firm 1 invests or not in the first stage. [25%]

Since firm 1 does not invest in stage 1 in the NE, from part (a) both firms' NE strategies must involve setting $q_i^* = 30$ if firm 1 does not invest.

Otherwise, the strategies cannot be mutual best responses.

Thus, $\pi_1^* = 900$ if firm 1 does not invest in the first stage.

Since firm 2 does not condition its production on whether firm 1 invests or not, she also must set $q_2^* = 30$ if firm 1 invests in the first stage.

NB: Since firm 1 does not invest in stage 1 in the NE, any production by firm 2 after investment is a BR.

Thus, since $q_1^* = \frac{96-q_2}{2}$ after investment from part (b), firm 1 cannot do better than setting $q_1^* = \frac{96-30}{2} = 33$ in the second stage after investing in the first stage which gives:

$$\pi_1^* = (96 - 33 - 30) \times 33 = 33^2 = 1089$$

But $1089 - 900 = 189 < 200 = K$, and so firm 1 does not wish to deviate from the NE and invest.

Since firm 1 does not condition its production on whether she invests or not, she must also set $q_1^* = 30$ if she invests in the first stage.

NB: Just like for firm 2, any production by firm 1 after investment is a BR.

Strategy for firm 1:

- Do not invest in stage 1.

If invested in stage 1, set $q_1^* = 30$ in stage 2;

if did not invest in stage 1, set $q_1^* = 30$ in stage 2.

Strategy for firm 2:

- If firm 1 invested in stage 1, set $q_2^* = 30$ in stage 2;

if firm 1 did not invest in stage 1, set $q_2^* = 30$ in stage 2.

- (f) Using your answers to parts (d) and (e), explain how subgame-perfect Nash equilibrium relates to the notion of credibility of agents' play in games. [12%]

Explain how the concept of SPNE ensures credibility by making sure that strategies are NE in every subgame and not just for the game as a whole.

Note that the SPNE in part (d) ensures the credibility of the quantity choices after firm 1's investment decision.

Explain that the strategies in part (e) are Nash since the strategies are best responses to each other.

Firm 2's strategy is not credible, because if firm 1 did invest the firms would play the equilibrium from part (b) in the second stage.

But Nash does not require that the firms' strategies be Nash in every subgame, and hence does not require that firm 2's quantity choice be credible when the firms set quantities after firm 1's invests off the equilibrium path.